

Problem

First fix

Better fix

In practice

Gradient descent
batch, stochastic,
mini-batch
Note 57–60

Weak spots of gradient descent

one learning rate,
saddle points
Note 1032

idea 1: build up speed

Momentum

EWMA of gradients
Note 1034

NAG

look ahead, then correct
Note 1035

Exponentially
weighted
moving average
Note 1033

AdaGrad

own learning rate
per parameter
Note 1036

RMSPProp

EWMA of
squared gradients
Note 1037

Adam

momentum +
RMSPProp
Note 1038

Keras Tuner

tries the
choices for us
Note 1039

A better
neural network
see the training map
Note 1021

idea 2: adapt the learning rate



fixes



used in



leads to

problem

from another map